

Supplemental document for filament exchange operation in 30kV RHEED e-gun : model AHR-100

1 introduction

This document describes recommended way to exchange the filament without breaking insulator parts

2 Preparation

- 1 Spare filament Type : D4-43651
- 2 Special tool Attached item
- 3 Hex wrench for M2 scr Attached item
- 4 Working desk which is suitable for operation.
- 5 Tweezer
- 6 Groove to avoid contamination for vacuum parts.
- 7 Alcohol and wiper for cleaning



3 Procedure

- 3 1 Stop pumping unit and break vacuum in the chamber.
HV connector Ring screw

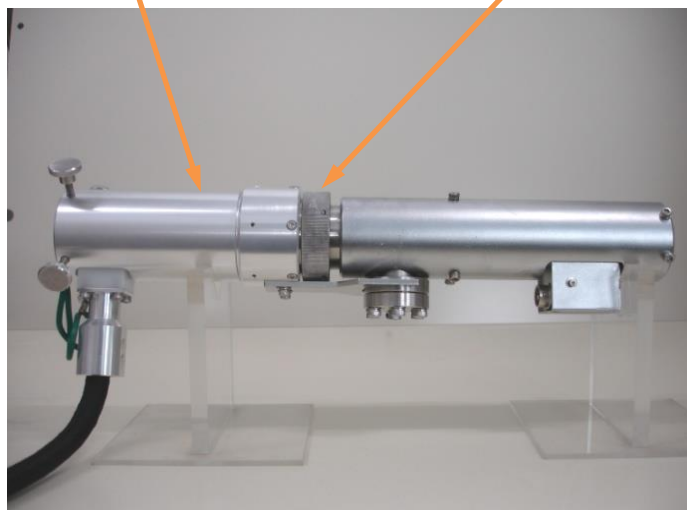


Fig-1 Overview of the RHEED e-gun

- 3 2 Loosen the ring screw (rotation direction in red) with mounting HV connector.

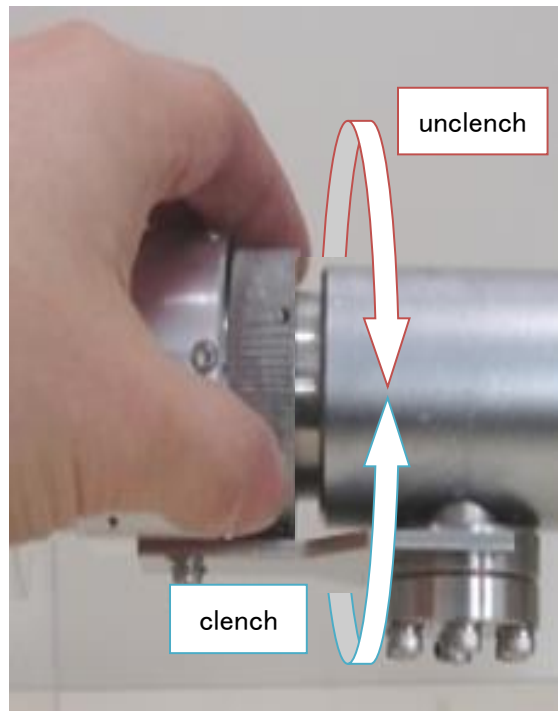
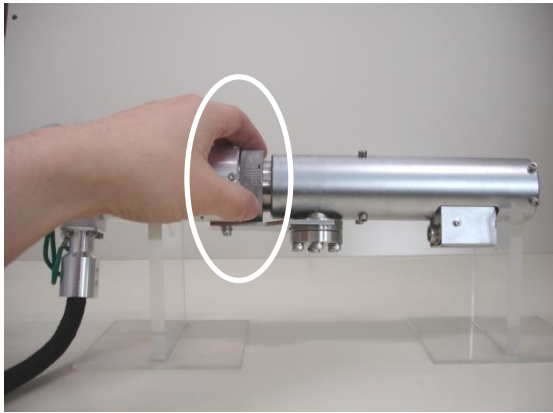


Fig-2 Rotation direction in ring screw (red: loosen and blue: fasten).

- 3 3 Remove the half part of the RHEED e-gun including filament/Wehnelt assembly and HV connector.

Whenelt
assembly

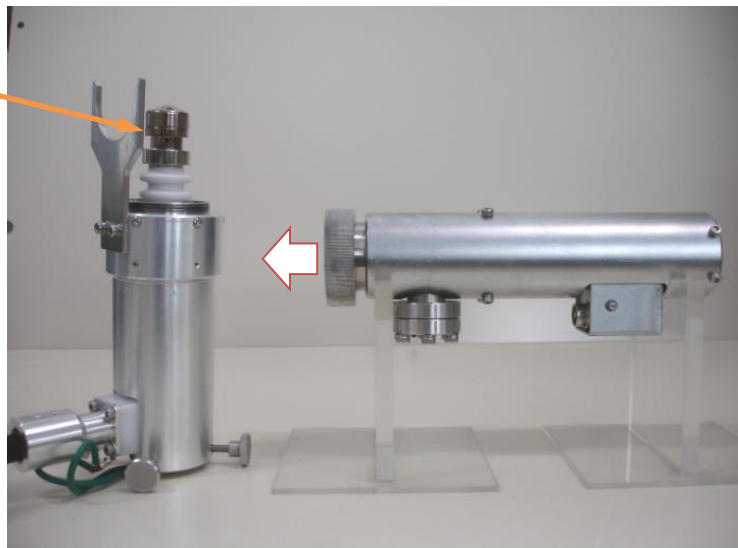


Fig-3 Remove filament/Wehnelt assembly and HV connector.

- 3 4 Set the special tool which checks height position of the Wehnelt electrode as shown in Fig 4.



Special tool
for filament exchange

Fig-4 Removed part and special tool.

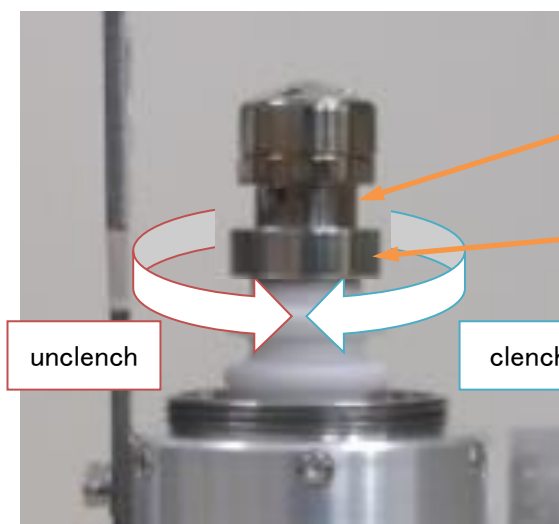


pay attention the HV cable works to prevent fall the part following the cable.

- 3 5 Remove the Wehnelt assembly with loosening lower ring screw without rotating upper part.



Keep fixing the rotation position of the Wehnelt assembly. It is possible to break the insulator if not keeping the position.



unclench

clench



Fig-5 Remove the Wehnelt assembly .

3 6 Set the Wehnelt part to the special tool.



Two pins in filament

Rectangle hole

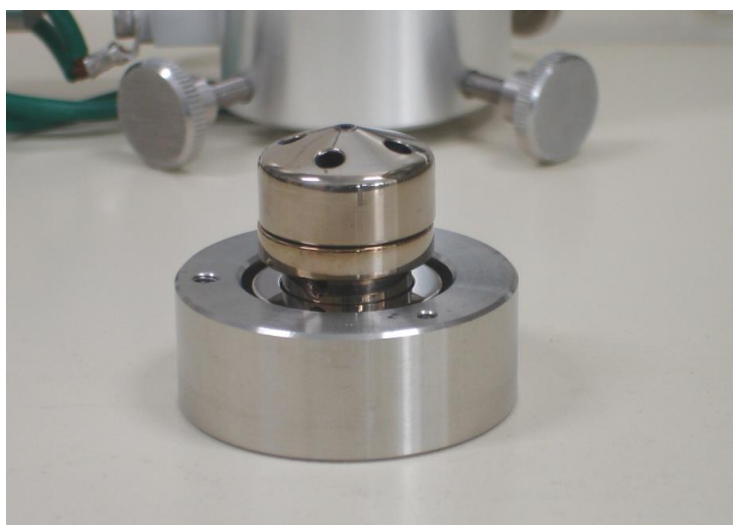
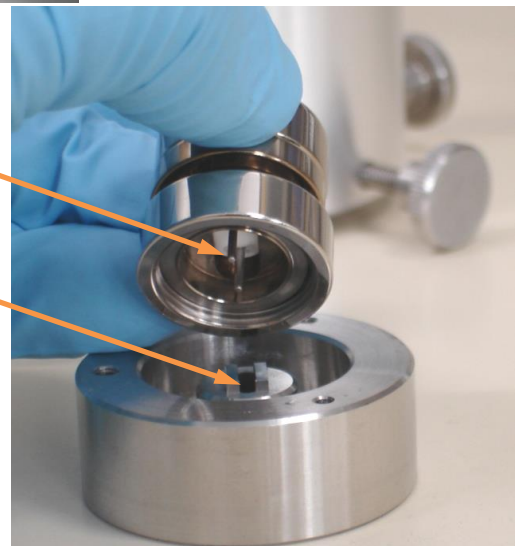


Fig-6 Setting sotation of the Wehnelt part on the specail tool.

- 3 7 Check the top position of the Wehnelt part with the special tool.

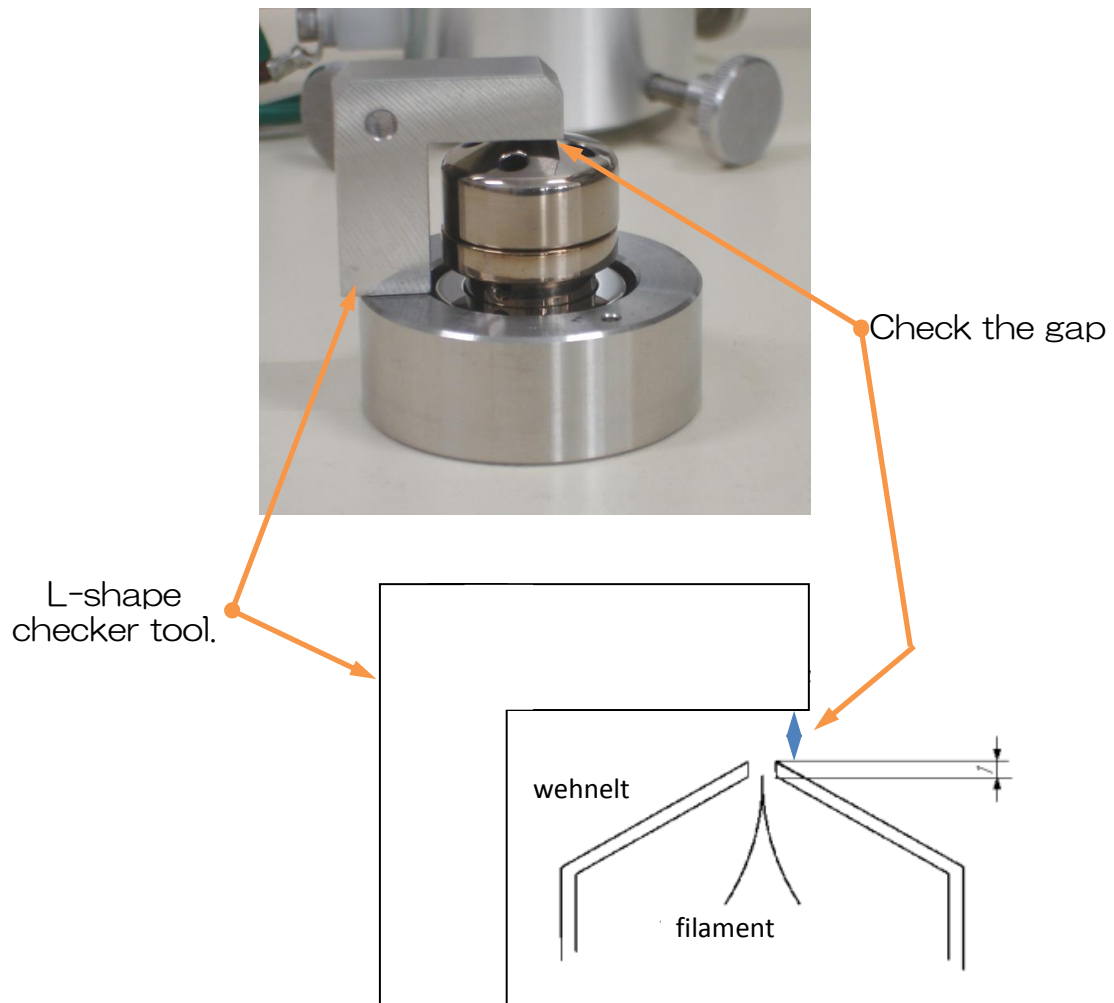


Fig-7 Spacial relationship between the Wehnelt and the filament.

- 3 8 Remopve the cap of the Wehenlt with loosing stopper screw.

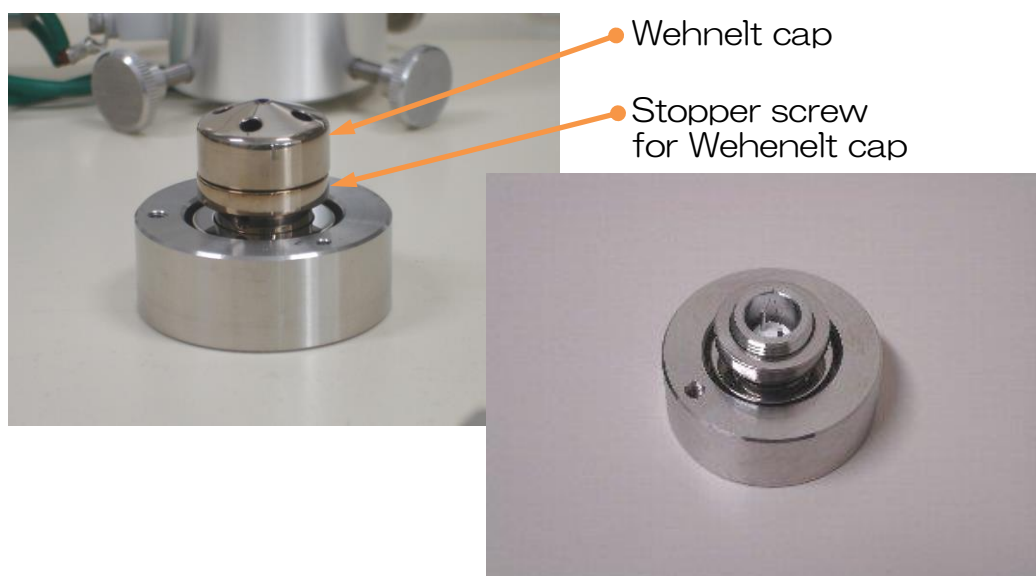


Fig-8 Image after removing Whenelt cap

- 3 9 Remove the old/broken filament with loosening side-fixing screw. It is good way to refer three marker at the special tool to find/arrange the screw position.



Reference marker for side fixing screws.

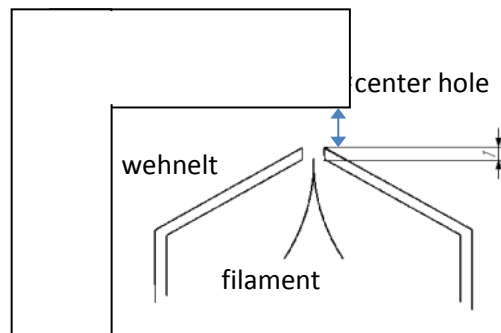
Mount the new filament following almost reverse order above.

Fig-9 Removing /mounting filament.



Casp of the filament easily breaks with touching. Handle the filament with tweezers at bottom side.

- 3 10 Set the height position of the Wehnelt cap which have been seen in former section 2.7 before removing. And check the horizontal position of the filament cap where the cap can be seen at center of the Wehnelt cap hole.

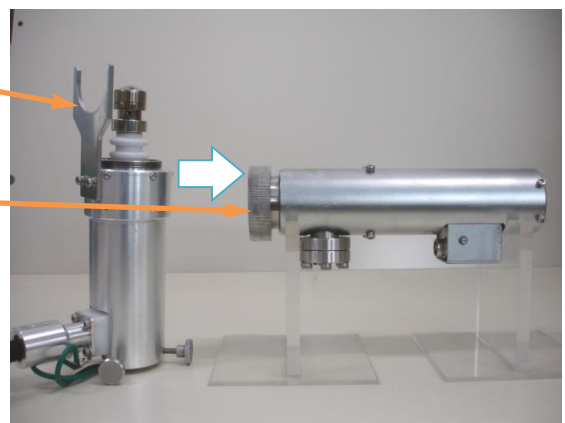


Do not rotate the Wehnelt part.

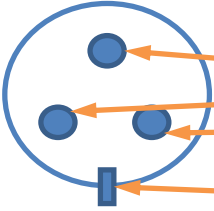
- 3 11 Mount the rear half part by reverse order of taking off.

Adjust the piping mount position.

Mount with cleaning O-ring elastomer to keep high vacuum.

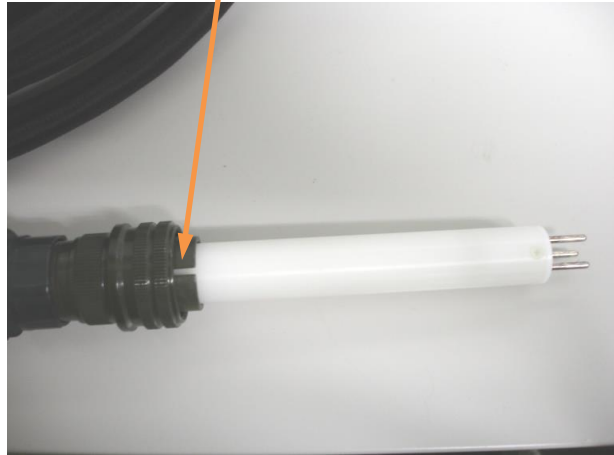


- 3 12 Remove the HV cable at opposite side on rear panel of HV power sup
And check the electric conduction.insulation.



	HV	Filament		ground
		1	2	
HV	-	X	X	X
Filament 1	X	-	O	X
Filament 2	X	O	-	X

Groove adjuste X : Insulate
O : Conductive



Mount ground cable
to HV connector

- 3 13 Recover the vacuum punping in the chamber.
check the e-gun performance.